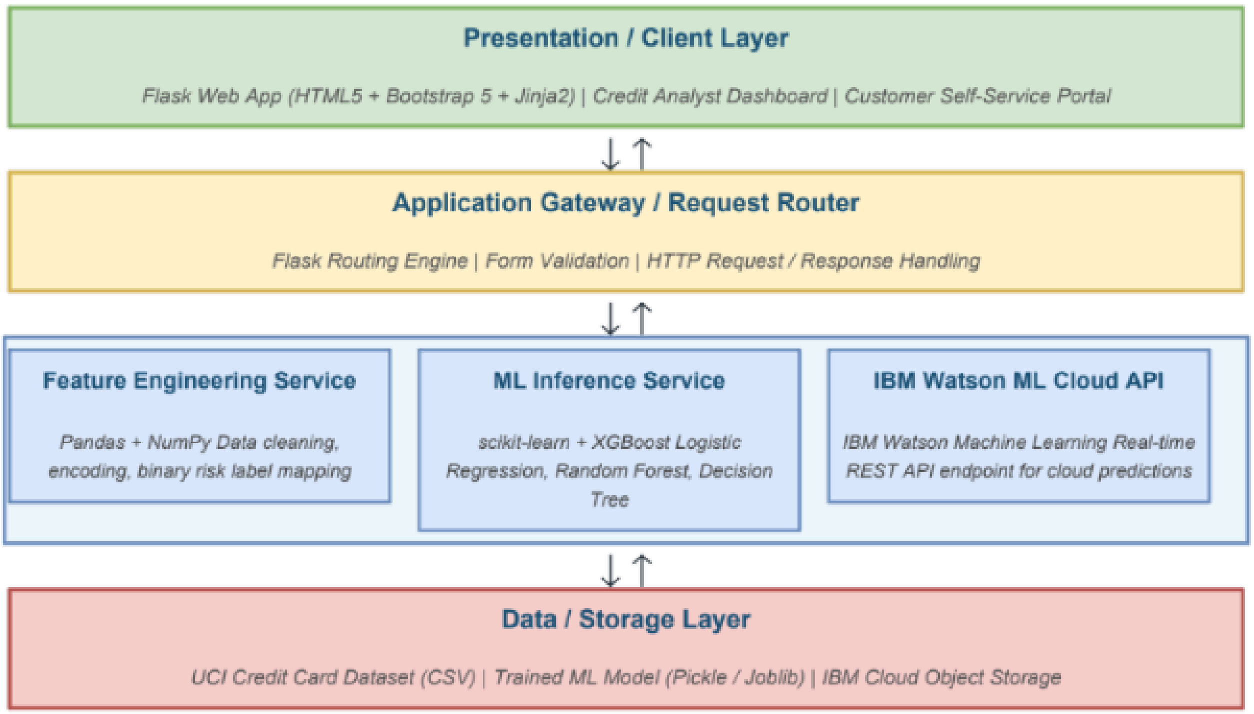
Solution Architecture

Date	30 june 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer (Frontend / UI)	Provides the web-based user interface for credit analysts and prospective customers. Users enter applicant financial data (income, employment, credit history) via HTML forms. Jinja2 templates dynamically render the ML prediction result (Approved / Rejected) returned by the Flask backend.	HTML5, CSS3, Bootstrap 5, Jinja2 Templates
Application Gateway (Flask Router)	Acts as the central request router and application entry point. Handles all HTTP GET/POST requests, performs server-side form validation, and routes data to the appropriate service (feature engineering pipeline or IBM Watson ML API) before returning the prediction response.	Python 3.10, Flask Framework, Flask Routing Engine
Feature Engineering Service	Pre-processes raw applicant input data before passing it to the ML model. Applies the same transformation pipeline used during model training: encoding categorical variables, scaling numerical features, and converting multi-class payment status codes into binary risk labels.	Python, Pandas, NumPy, scikit-learn Preprocessing Pipeline
ML Inference Service (Core Logic)	Hosts the best-performing trained classification model (selected from Logistic Regression, Random Forest, XGBoost, Decision Tree based on accuracy and F1-score). Receives preprocessed feature vectors and returns a binary approval prediction (Approved / Rejected).	scikit-learn, XGBoost, Pickle / Joblib

IBM Watson ML Cloud API	Provides a scalable, cloud-hosted REST API endpoint for the deployed ML model on IBM Cloud. Enables real-time predictions accessible from any client without requiring local model installation. Supports future scaling and enterprise-grade SLA requirements.	IBM Watson MachineLearning, IBMCloud, REST API
Data / Storage Layer	Stores the historical applicant dataset used for model training and evaluation. The trained model artifact is serialised and stored as a Pickle/Joblib file, loaded at Flask startup. IBM Cloud Object Storage is used for cloud-side model artifact management.	UCI Credit Card Dataset (CSV), Pickle / Joblib, IBM Cloud Storage
Version Control & Development Tools	GitHub is used for source code versioning, team collaboration, and tracking all changes to the ML pipeline and Flask application. Jupyter Notebook is used for exploratory data analysis, model training, and evaluation. Postman is used to test Flask and Watson API endpoints.	GitHub, Jupyter Notebook, Postman